

# Part 16: Bayesian Estimation

## Summary

To this point, our discussion has focused on the **frequentist** or **classical** approach to estimation and hypothesis testing. In that framework, consideration is given to how procedures perform in repeated use. Indeed, the performance of estimators can be assessed without using any actual data. Some find this troubling.

An alternative philosophy is the **Bayesian** approach to estimation. The central characteristic is that uncertainty in parameters is quantified using probability distributions.

As the name implies, Bayes' Rule plays a central role.

First, recall Bayes' Rule: If  $B_1, B_2, \dots, B_m$  are a partition of  $\Omega$ , then

$$P(B_i|A) = P(A|B_i)P(B_i) / \sum_j P(A|B_j)P(B_j)$$

Also recall the continuous version:

$$f_{Y|X}(y|x) = f_{X|Y}(x|y)f_Y(y) / \int f_{X|Y}(x|u)f_Y(u) du$$

Here we will use this result with the parameter  $\theta$  "playing the role" of  $Y$ :

$$\pi(\theta|\mathbf{x}) = f(\mathbf{x}|\theta)\pi(\theta) / \int f(\mathbf{x}|t)\pi(t) dt$$

The vector  $\mathbf{x}$  is observed data.

The density  $\pi(\theta)$  is the **prior distribution** for  $\theta$ .

The conditional density  $\pi(\theta|\mathbf{x})$  is the **posterior distribution** for  $\theta$ .

- 1 In the Bayesian framework, the prior reflects the someone's **opinion** re-
- 2 garding the value of  $\theta$ , **prior to** the observation of the data.
- 3 The posterior is this opinion "updated" to take the data into consideration.
- 4 *How is the prior chosen?* This can be tricky, and subjective.
- 5 The use of the Bayesian approach was long limited by computational diffi-
- 6 culties associated with calculating the required integral. With the advent of
- 7 **Markov Chain Monte Carlo (MCMC)** algorithms, these issues have largely
- 8 gone away, and the use of Bayesian methods has grown greatly.
- 9 In some relatively simple cases, analytical solutions are possible.

- 1 **Exercise:** Suppose that the observable data are modelled by i.i.d.
- 2 Bernoulli( $\theta$ ) random variables  $X_1, X_2, \dots, X_n$ . You quantify your prior
- 3 beliefs regarding  $\theta$  by using the Beta( $\alpha, \beta$ ) distribution, for some chosen
- 4 values of  $\alpha$  and  $\beta$ . Derive the posterior for  $\theta$  in this case.

5 First, recall the Beta dist:

$$6 \quad \pi(\theta) = \left( \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \right) \theta^{\alpha-1} (1-\theta)^{\beta-1} \quad 0 < \theta < 1$$

8 The dist for data  $f(\underline{x}|\theta)$

$$9 \quad f(\underline{x}|\theta) = \prod_{i=1}^n f(x_i|\theta) = \prod_{i=1}^n \theta^{x_i} (1-\theta)^{1-x_i}$$

$$10 \quad = \theta^{\sum x_i} (1-\theta)^{n - \sum x_i}$$

$$11 \quad = \theta^{\sum x_i} (1-\theta)^{n - \sum x_i}$$

$$12 \quad = \theta^{\sum x_i} (1-\theta)^{n - \sum x_i}$$

Now use Bayes Rule

$$\pi(\theta | \underline{x}) = \frac{f(\underline{x} | \theta) \pi(\theta)}{\int f(\underline{x} | t) \pi(t) dt} = \underline{k_1} f(\underline{x} | \theta) \pi(\theta)$$

where  $k_1$  is not a function of  $\theta$

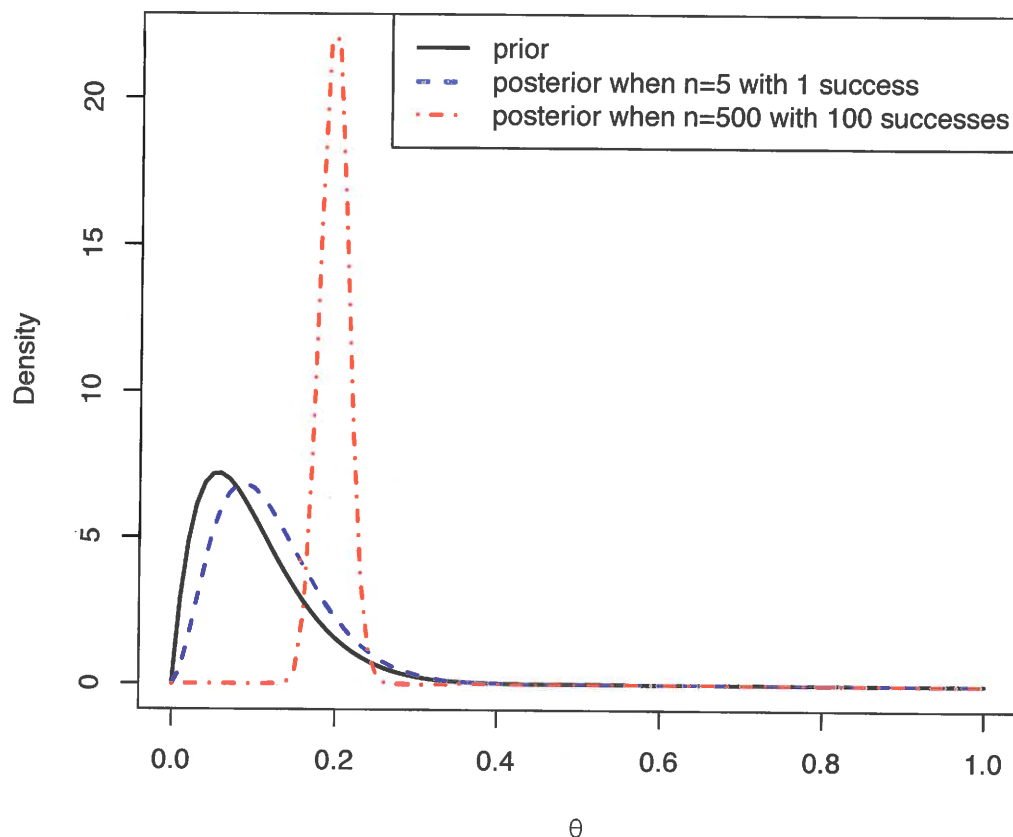
$$= \left[ \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} \right] \left[ \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1} \right]$$

$$= k_2 \theta^{(\alpha-1)+\sum x_i} (1-\theta)^{(\beta-1)+(n-\sum x_i)} \quad , \quad 0 < \theta < 1$$

where  $k_2$  does not depend on  $\theta$

So, that this is the Beta( $\alpha + \sum x_i$ ,  $\beta + (n - \sum x_i)$ ) distribution.

- 1 In this example, we see that the prior and posterior are from the same
- 2 family of distributions (i.e., they are both Beta). We say that “The Beta
- 3 distribution is conjugate to sampling from the Bernoulli.”
- 4 Let’s consider a couple specific examples. Suppose I chose  $\alpha = 2$  and
- 5  $\beta = 18$  to capture my prior opinion regarding  $\theta$ . I observe  $n = 5$  trials, and
- 6  $\sum_i x_i = 1$ . The result above states that the posterior is Beta( $2+1, 18+(5-1)$ ).
- 7 If instead,  $n = 500$  and  $\sum_i = 100$ , the posterior is Beta( $102, 418$ ).
- 8 The figures on the following slide compare these cases.



- 1 **Exercise:** Comment on the shapes of the posteriors in the previous exam-  
 2 ple, as compared to the prior. What is the role of the sample size?

3 *With small sample, the posterior is similar to*  
 4 *the prior. As sample size grows, the*  
 5 *data "swamp" the prior.*

- 1 The Bayesian often wants to convert the posterior into an estimate for  $\theta$ .  
 2 One way to do this is to find the maximum of the posterior, called the  
 3 **maximum a posteriori (MAP) estimator**.

- 4 In our example this comes out to be

5 
$$\frac{\alpha + \sum x_i - 1}{\alpha + \beta + n - 4}$$

*mode of Beta( $\alpha + \sum x_i$ ,  $\beta + (n - \sum x_i)$ )*

- 6 Another approach is to find the posterior mean, called the **Bayes estimator**.

- 7 In our example the Bayes estimator is

8 
$$E(\theta|\mathbf{x}) = \frac{\alpha + \sum x_i}{\alpha + \beta + n} = \left( \frac{\sum x_i}{n} \right) \left( \frac{n}{\alpha + \beta + n} \right) + \left( \frac{\alpha}{\alpha + \beta} \right) \left( \frac{\alpha + \beta}{\alpha + \beta + n} \right)$$

- 9 In other words, it is a weighted average of the sample proportion and the  
 10 prior mean. The weight placed on the sample proportion grows with  $n$ .

- 1 **Exercise:** Return to our Bernoulli example. Choose  $\alpha = \beta = 1$  for the prior.  
 2 What is the appeal of this choice? What is the Bayes estimator?

3 If  $\alpha = \beta = 1$ , the prior is  $\text{Uniform}(0, 1)$   
 4 "flat prior"  $\Rightarrow$  "all values of  $\theta$  equally likely"

$$E(\theta | \underline{x}) = \left( \frac{\sum x_i}{n} \right) \left( \frac{n}{n+2} \right) + \left( \frac{1}{2} \right) \left( \frac{2}{n+2} \right)$$

$$\neq \left( \frac{\sum x_i}{n} \right)$$

10 "There is no uninformative prior"

- 1 **Exercise:** The observable data are modelled using the  $\text{Poisson}(\theta)$  distribu-  
 2 tion. Use the  $\text{Gamma}(\alpha, \beta)$  distribution as the prior on  $\theta$ . Find the posterior  
 3 distribution and the Bayes estimator.

4  $X_1, X_2, \dots, X_n$  iid  $\text{Poisson}(\theta) \Rightarrow f(x_i | \theta) = \frac{e^{-\theta} \theta^{x_i}}{x_i!}$

5  $f(\underline{x} | \theta) = \prod_{i=1}^n f(x_i | \theta) = \dots$

7  $\pi(\theta) =$

9  $\pi(\theta | \underline{x}) = \frac{f(\underline{x} | \theta) \pi(\theta)}{\int f(\underline{x} | t) \pi(t) dt}$

12 Posterior is  $\text{Gamma}(\alpha + \sum x_i, \beta + n)$

### Exercise

$X_1, X_2, \dots, X_n$  iid  $\text{Poisson}(\theta)$

$$\text{Hence } f(x_i|\theta) = \frac{e^{-\theta} \theta^{x_i}}{x_i!} \text{ for } i=1, 2, \dots, n$$

and joint distribution is

$$f(\underline{x}|\theta) = \frac{e^{-n\theta} \theta^{\sum x_i}}{\prod_{i=1}^n x_i!}.$$

We are assuming a  $\text{Gamma}(\alpha, \beta)$  prior for  $\theta$ ,

$$\text{so } \pi(\theta) = \frac{\theta^{\alpha-1} e^{-\beta\theta} \beta^\alpha}{\Gamma(\alpha)}, \quad \theta > 0.$$

The posterior is

$$\pi(\theta|\underline{x}) = \frac{f(\underline{x}|\theta) \pi(\theta)}{\int f(\underline{x}|\theta) \pi(\theta) d\theta}$$

$$= k_1 f(\underline{x}|\theta) \pi(\theta)$$

where  $k_1$  does not depend on  $\theta$

$$= k_1 \left( \frac{e^{-n\theta} \theta^{\sum x_i}}{\prod_{i=1}^n x_i!} \right) \left( \frac{\theta^{\alpha-1} e^{-\beta\theta} \beta^\alpha}{\Gamma(\alpha)} \right), \theta > 0$$

$$= k_2 e^{-n\theta} \theta^{\sum x_i} \theta^{\alpha-1} e^{-\beta\theta}, \theta > 0$$

where  $k_2$  does not depend on  $\theta$

$$= k_2 e^{-(n+\beta)\theta} \theta^{(\alpha-1)+\sum x_i}, \theta > 0$$

This is the  $\text{Gamma}(\alpha + \sum x_i, \beta + n)$  distribution.

Hence, the posterior is  $\text{Gamma}(\alpha + \sum x_i, \beta + n)$ .

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## 1 Higher Dimensional Parameters

- 2 The Bayesian approach can be extended to multi-dimensional parameters.
- 3 The difficulties are more computational than conceptual. It is possible,
- 4 for instance, to place prior distributions on covariance matrices with tools
- 5 tailored to this purpose. Choosing priors can be even trickier in higher
- 6 dimensions, however.  $\pi(\Sigma)$
- 7 For example, the paper "Bayesian Portfolio Analysis" by Avramov and
- 8 Zhou (*Ann. Rev. of Financial Economics*, Vol. 2) provides a nice overview of
- 9 Bayesian tools for selecting portfolios. Consider Equation (8) for the un-
- 10 derlying optimization problem. Note how uncertainty in the mean vector
- 11  $\mu$  and covariance matrix  $V$  are taken into consideration: Fixed estimates
- 12 are not plugged in, instead one integrates over the posterior for these quan-
- 13 tities.

## 1 Numerical Approach in R

- 2 One of the challenges of doing Bayesian inference is the (sometimes) com-  
3 putational difficulty of calculating the denominator of the fraction

$$4 \quad \pi(\theta|\mathbf{x}) = f(\mathbf{x}|\theta)\pi(\theta) / \int f(\mathbf{x}|t)\pi(t) dt$$

- 5 In “nice” cases we have a conjugate prior, and we can hope that that class  
6 of distributions is sufficiently broad to capture our prior “opinion” regard-  
7 ing  $\theta$ .

- 8 In cases where  $\theta$  is low dimensional, numerical integration is practical, and  
9 the `integrate()` function can be very helpful.

- 1 The `integrate()` function works much in the same way as `optim()`,  
2 except instead of numerically minimizing a function, it will numerically  
3 integrate a function.

- 4 As with `optim()`, the first argument to `integrate()` is another R func-  
5 tion, the function which is to be integrated. The first of argument of this  
6 second function must be the “ $x$ ” variable over which the integral will be  
7 calculated. You set the range of integration using the `lower` and `upper`  
8 arguments.

- 9 For example,

```
> integrate(dnorm, lower=0, upper=1)
```

```
0.3413447 with absolute error < 3.8e-15
```

$P(0 < Z < 1)$

1 A few things to note:

2 1. It is fine to use `lower = -Inf` and/or `upper = Inf`.

3 2. R has the (annoying) habit of returning the “with absolute error” bit  
4 as the default. You can use instead

```
> integrate(dnorm, lower=0, upper=1)$value  
[1] 0.3413447
```

5 3. The function to be integrated must be “vectorized.” This means that  
6 if a vector of  $x$  values are passed into  $f(x)$ , then that function must  
7 return a vector of equal length. This, of course, works for `dnorm`:

```
> dnorm(c(1, 2, 3))  
[1] 0.241970725 0.053990967 0.004431848
```

1 4. If you want to pass additional arguments on to the function being inte-  
2 grated, then specify these, with their names, as additional arguments  
3 to `integrate()`. For example,

```
> integrate(dnorm, lower=-Inf, upper=0, mean=1, sd=2)  
0.3085375 with absolute error < 4.8e-07
```

- 1 Let's demonstrate this with a simple example.
- 2 Suppose our model is that  $X_1, X_2, \dots, X_n$  are i.i.d.  $\text{Normal}(\mu, \sigma^2 = 2)$ . Our
- 3 "prior opinion" is that  $\mu$  is equally likely to be anything between 1 and 5.
- 4 We observe the following vector of data
 

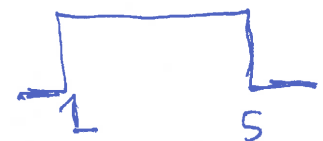
```
> dat = c(0.44, 0.92, 1.81, 0.22, 1.04, 0.38, 2.76,
+         1.02, 3.07, 1.45)
```
- 5 What is the posterior for  $\mu$  given this data?

- 1 So, we can construct a function as follows to calculate the posterior:

```
> posterior = function(mu, dat)
+ {
+   numerator = function(mu, dat)
+   {
+     out = dunif(mu, 1, 5)
+     for(i in 1:length(dat))
+     {
+       out = out * dnorm(dat[i], mu, sqrt(2))
+     }
+     return(out)
+   }
+   denominator = integrate(numerator, lower = 0,
+     upper = 6, dat=dat)$value
+   return(numerator(mu, dat)/denominator)
+ }
```

$$\pi(\theta | \tilde{x}) = \frac{f(\tilde{x} | \theta) \pi(\theta)}{\int f(\tilde{x} | t) \pi(t) dt}$$

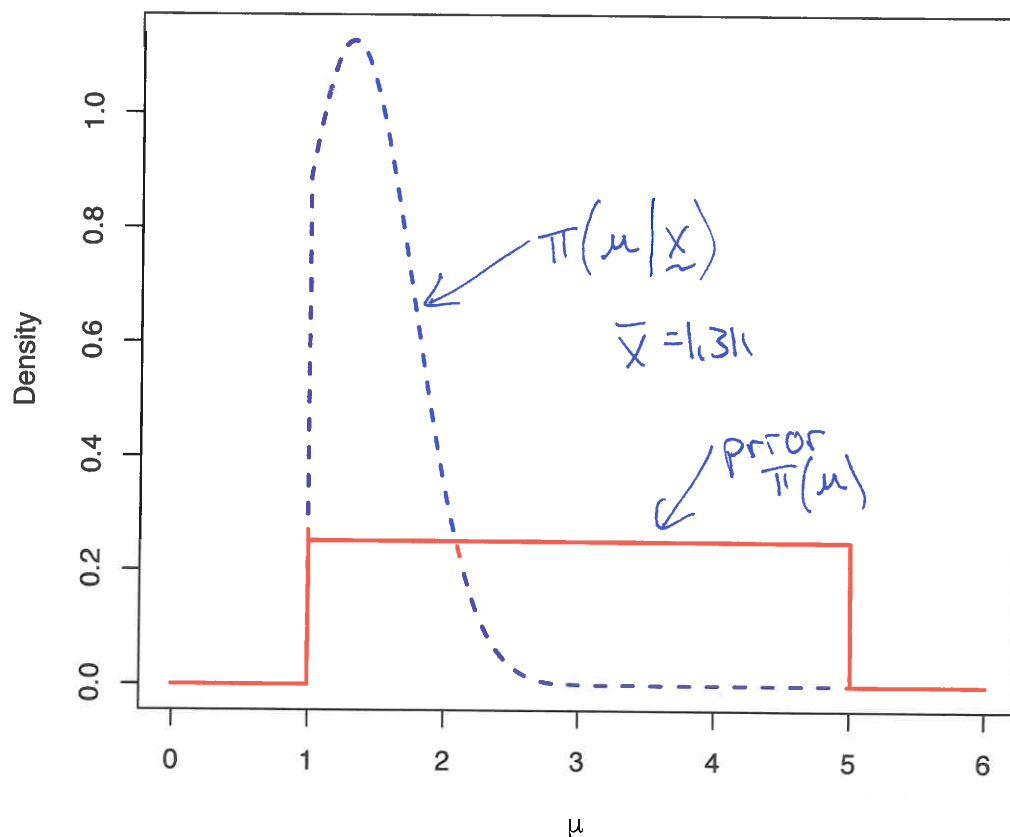
$$f(x_i | \mu)$$



- 1 Note that the function `numerator` is “vectorized” in terms of `mu`, allowing
- 2 it to be integrated using the `integrate()` function.
- 3 Also, we note that the function `posterior()` is also vectorized in terms
- 4 of `mu`, so that I can use the `curve()` function with it:

```
> curve(posterior(x,dat), from = 0, to = 6,
+       xlab=expression(mu), ylab="Density",
+       col=4, lwd=2, lty=2, n=1000)
```

- 5 This posterior is shown in the figure below, along with the uniform prior.



- 1 We can calculate the MAP estimate:

```
> negposterior = function(x,...) { (-1)*posterior(x,...) }  
> optim(3, negposterior, dat=dat, method="BFGS")$par  
[1] 1.311
```

- 2 We can find the Bayes estimator numerically as well:

```
> posterior_times_mu = function(mu,dat)  
+ {  
+   return(mu * posterior(mu,dat))  
+ }  
> integrate(posterior_times_mu,lower=-Inf,upper=Inf,dat=dat)$value  
[1] 1.431109
```

- 3 Compare these both with the sample mean:

```
> mean(dat)  
[1] 1.311
```

## Bootstrap

$$X_1, X_2, \dots, X_n$$

$T(X_1, X_2, \dots, X_n)$  statistic, e.g.  $\hat{\theta}$  an estimator

With bootstrap, resample the data, i.e. draw with replacement a sample of size  $n$  from your original sample

$$X_1^*, X_2^*, \dots, X_n^*$$

then calculate  $T(X_1^*, X_2^*, \dots, X_n^*)$ . Store as  $T_1^*$

Repeat many times, get  $T_1^*, T_2^*, \dots, T_B^*$   
↑  
# of bootstrap reps

dist of  $T_1^*, T_2^*, \dots, T_B^* \approx \text{dist. of } T$

$$\text{e.g. } V(T_1^*, T_2^*, \dots, T_B^*) \approx V(T)$$

$$SE(\quad) \approx SE(T)$$